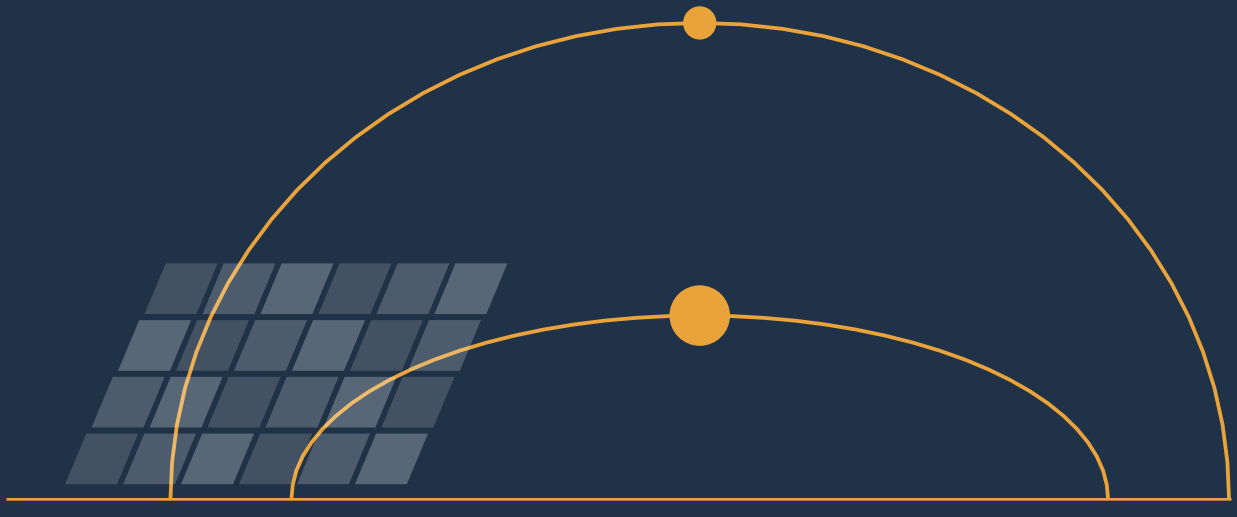


LARCH HILL TECHNICAL NOTES · NO. 4

Power for a Cabin

Sizing a small off-grid solar system

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SECOND EDITION

SIZING A SMALL OFF-GRID SOLAR SYSTEM

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Larch Hill Technical Notes are short handbooks for people who build and maintain their own equipment. The cabin, the valley and the figures in this one are an example worked by hand: check every number against your own loads, site and suppliers before you buy anything.

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CHAPTER

1 The daily load

Every component of an off-grid system is sized from one number: the energy the cabin uses on a winter day. Get it wrong and nothing downstream can put it right.

A solar system for a cabin is sized backwards. You start at the sockets, add up what the cabin draws in a day, and only then work out how many panels and batteries it takes to supply that much in the worst month of the year. This chapter arrives at that number: 1,159 watt-hours a day, which the rest of the book rounds to 1,160. **Chapter 2** turns it into panels, and **chapter 3** into batteries.

The example is a stone cabin of 48 m² at 1,100 m in a valley that faces south-east, used every weekend and for three weeks in winter. Heating and cooking are wood and bottled gas; electricity runs the lights, a fridge, a water pump, a laptop and a router.

1.1 Listing the loads

Walk round the cabin with a notebook and write down everything that plugs in or is wired in, with its power in watts and the hours it runs on a winter day. The power is on the rating plate or in the manual; for anything with a motor or a compressor, a plug-in meter left for a day gives a truer figure than the plate. **Table 1.1** is the list for the example cabin.

Two lines in the table are easy to forget. The inverter, which turns the battery's 24 volts into 230 volts for the sockets, draws 8 W from the moment it is switched on, whether anything is plugged in or not: over a day that is more than the lights. The router runs all night as well. Both are worth a timer or a switch by the door, and the saving is worked out in **section 1.3**.

The fridge figure needs care. A chest fridge of 100 litres draws 55 W while its compressor runs, and in a cool cabin the compressor runs about five hours in twenty-four. In August it may run nine, but August is not the month that sizes the system, as **section 2.1** shows on **p. 6**.

Table 1.1. Daily loads of the example cabin on a winter day

Load	Power (W)	Hours a day	Wh a day
LED lighting, 6 points	30	5	150
Chest fridge, 100 L	55	5	275
Water pump, 24 V	120	0.5	60
Laptop	45	4	180
Router	8	24	192
Inverter, idle draw	8	24	192
Two phones	10	2	20
Wood-stove fan	15	6	90
Total			**1,159**

1.2 When the energy is used

The total says how much energy the cabin needs; it does not say when. **Fig. 1.1** spreads the same 1,160 Wh over the hours of a winter day. The fridge, the router and the inverter make a floor of about 27 W that never goes away. The laptop adds a block in the morning, and the evening brings the largest draw of the day, when lights, the stove fan and the pump run together after sunset.

That evening peak matters more than its size suggests. Almost all of it falls after the sun has gone, so none of it can come straight from the panels: it is drawn from the battery and paid back the next day. The battery bank in **section 3.1** is sized for exactly this, and for the days when the sun does not pay it back at all.

1.3 The winter margin

A list made in October is a guess about January. Lights run longer in midwinter, guests come at New Year, and someone always brings a hair dryer. Rather than pad every line of **table 1.1**, keep the list honest and add the margin once, at the end, where it can be seen.

For a weekend cabin a margin of 15% is enough, and the example takes it from the loads themselves rather than adding it on top: switching the inverter and the router off at night saves 8 W and 8 W for ten hours, 160 Wh a day, close

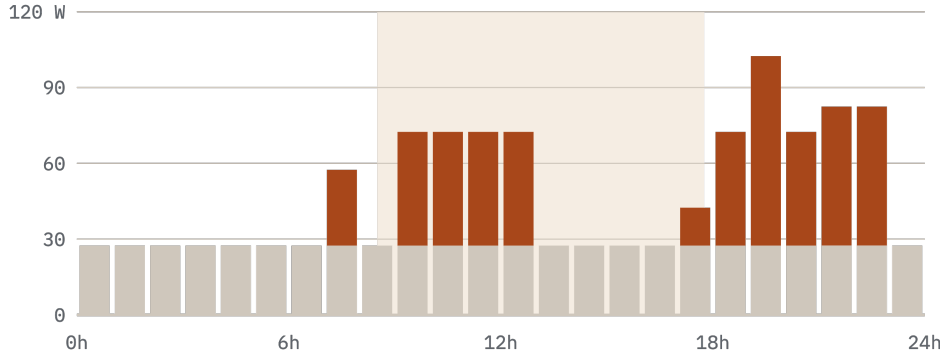


Figure 1.1. Average draw, hour by hour, on a winter day. Grey: the constant floor of fridge, router and inverter; orange: everything else; pale band: daylight.

to 14% of the total. The system is sized for the full 1,160 Wh, and the habit of the switch by the door is the margin.

With the load fixed, the next question is how much of it the sun can supply in the darkest month. That is the work of **chapter 2**, which starts from the sun hours of the site on **p. 6**.

CHAPTER

2 The array

Panels are sized for December, when the days are short and the sun is low. In summer the same array makes twice what the cabin needs, and that is the price of a winter that works.

Chapter 1 ended with a daily load of 1,160 Wh, the total of **table 1.1**. This chapter finds the array that delivers that much on an average December day, after every loss between the panel and the socket.

2.1 Sun hours by month

The sunlight a site receives is given in peak sun hours: the number of hours at a standard intensity of 1,000 W/m² that would deliver the same energy as the whole day does. A panel rated at 300 W makes roughly 300 Wh for each peak sun hour, before losses. The figures come from a solar database for the site's coordinates and the panels' tilt; **fig. 2.1** gives them for the example cabin, at 42° N with the panels tilted at 60°.

A steep tilt trades some summer yield for the low winter sun. Even so, December gets half the yield of July: 2.8 peak sun hours against 5.6. December is therefore the design month, and every calculation in this chapter uses its figure.

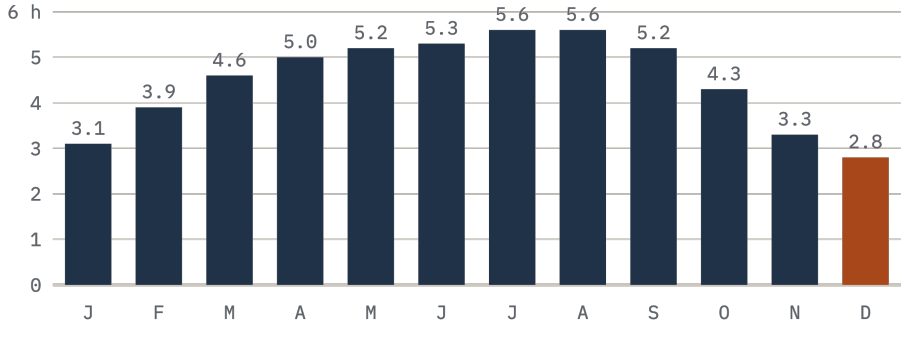


Figure 2.1. Peak sun hours a day by month at 42° N, panels tilted at 60°. December is the design month.

2.2 Losses between panel and socket

Not all of the energy a panel makes reaches the sockets. Each stage between them takes a share, and the shares multiply. **Table 2.1** lists them for the example system.

The largest single loss is the inverter's, and it applies only to the loads that run on 230 V. In a cabin where the fridge, the pump and the lights all run on 24 V, the overall factor rises from 0.79 to about 0.85. The example keeps the inverter in the chain, because the laptop and the router need it and the calculation is safer that way. The battery's share is its round-trip loss, the energy that goes in and does not come out; it is small for lithium cells and larger for lead-acid, as **section 3.2** explains on **p. 8**.

2.3 Sizing the array

The array must make the daily load, divided by the overall loss factor, in the peak sun hours of December:

```
1,160 Wh ÷ (2.8 h × 0.79) = 524 W
```

The example uses two panels of 310 W in series, 620 W in all, which leaves a margin of 18% for an old panel, a cloudy fortnight or a load that has grown.

The panels work between ten and four; the cabin, as **fig. 1.1** shows, spends most of its energy after dark. How large the battery between them must be is the subject of **chapter 3**.

Table 2.1. Losses between the panels and the sockets, multiplied

Between panel and socket	Factor
Dust, snow and shading	0.95
Cable resistance	0.97
MPPT charge controller	0.97
Battery round trip (LiFePO4)	0.95
Inverter, 230 V loads	0.93
Overall (product)	**0.79**

CHAPTER

3 The battery bank

The battery carries the cabin through every night and through the grey days when the panels make almost nothing. Its size is a choice about how many such days in a row you will accept.

The array of **section 2.3** works between ten and four, the cabin mostly after dark, and on an overcast December day the 2.8 peak sun hours of **fig. 2.1** fall to half an hour or less. The battery bank bridges both gaps.

3.1 Days of autonomy

Autonomy is the number of days the bank can run the cabin with no sun at all. Three days is usual for a cabin in a valley with fog, and it is what the example uses:

```
3 days × 1,160 Wh = 3,480 Wh usable
```

The daily load is the total of **table 1.1**, without the margin of **section 1.3** (**p. 4**), which stays in reserve. How far a bank may be drawn down depends on its chemistry.

3.2 Choosing a chemistry

Table 3.1 compares the three chemistries on sale for small off-grid systems. Lithium iron phosphate (LiFePO4) cells can be drawn down to 20% every night for thousands of cycles. Lead-acid batteries last longest when they are never taken below half their charge, so the same usable energy needs a bank about 60% larger, and five times the weight to carry up the track.

The round trip matters too. The overall factor of 0.79 in **table 2.1** assumes the lithium bank's 0.95. With an AGM bank at 0.85 the factor falls to 0.71, and the formula of **section 2.3** asks for 586 W instead of 524 W. The two 310 W panels still cover it, with a margin of 6% instead of 18%. Change the battery, and tables **2.1** and **3.1** must be read again together.

The example cabin uses a LiFePO4 bank of 200 Ah at 25.6 V: 5.1 kWh nominal, 4.1 kWh usable, a little more than three days.

Table 3.1. Three battery chemistries for 3,480 Wh of usable energy

Chemistry	Usable depth	Round trip	Cycles	Bank for 3 days	Mass
LiFePO4	80%	0.95	4,000	200 Ah, 5.1 kWh	45 kg
AGM lead-acid	50%	0.85	600	300 Ah, 7.2 kWh	220 kg
Tubular lead-acid	50%	0.80	1,500	300 Ah, 7.2 kWh	260 kg

3.3 A week of cloud

Fig. 3.1 follows the bank's state of charge through five December days: one clear, three overcast at about half a peak sun hour, and a clear day to finish. The bank starts at 80%, rises to 94% on the first afternoon and then loses about 17% a day, with a dip each evening from the peak of **fig. 1.1**. At its lowest, on the last night of cloud, it holds 27%, above the 20% floor.

The figure also shows the weakness of sizing for the average December day. On a clear day the array makes about 1,370 Wh after losses, only 210 Wh more than the cabin uses, so a bank drawn down by three grey days takes some twelve clear days to fill again. A third panel, on a controller of its own, raises the surplus of a clear day from 210 Wh to about 900 Wh; a small petrol generator with a charger refills the bank in an afternoon.

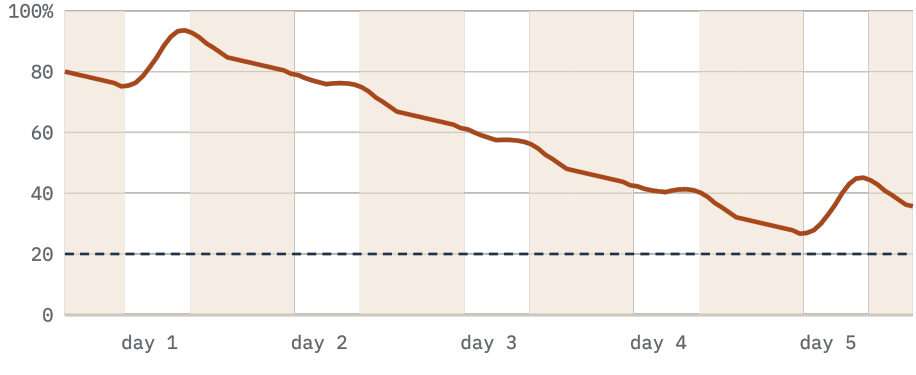


Figure 3.1. State of charge of the 5.1 kWh bank over five December days: clear, three overcast, clear. Shaded: night. Dashed: the 20% floor.